

Cross-Firm Information Flows and the Predictability of Stock Returns

ABSTRACT

We identify all return leader-follower pairs among individual stocks using Granger causality regressions. Thus-identified leaders can reliably predict their followers' returns out of sample, and the return predictability works at the level of individual stocks rather than industries. Our results indicate that, independent of its size, any firm may emerge as a return leader by being at the center of an important news development that has ramifications for other firms. Indeed, stocks undergoing news-generating developments see an increase in the number of stocks whose returns they lead.

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I. Introduction

In early 1994, six African-American employees of Texaco Inc. filed a racial discrimination lawsuit against their employer claiming that they were discriminated against in salaries and promotions. In an attempt to expedite a resolution, Reverend Jesse Jackson called for a national boycott of Texaco Inc. The lawsuit was eventually settled in late 1996 for over \$140 million, making it the largest settlement for a racial discrimination case at the time. As described in a New York Times article on November 17, 1996, the lawsuit potentially affected other

companies as well. In particular, Rev. Jackson announced not only that the Texaco boycott would continue but also that his organization, the Rainbow PUSH Action Network, would study the affirmative action policies of other companies that shared directors with Texaco Inc., such as Gillette, Johnson & Johnson, and Campbell Soup. The article also quoted a lawyer representing firms in discrimination lawsuits as saying, "If you are a consumer-product company, you are quite vulnerable. If you're an Exxon, or an American Express, or a Texaco, it's a big exposure."

While prior literature has shown that a stock with a high level of investor attention can lead returns of stocks with low levels of investor attention by being the first to react to common market- or industry-wide news, the evidence in this paper suggests that a stock can also lead returns of other stocks by being at the center of a news development that has ramifications for other firms. There are many instances in which firm-specific news could affect other companies. For example, the discovery of questionable accounting practices at one firm may cause investors to lose faith in financial statements of other firms that apply similar accounting techniques. Labor scandals or product safety concerns may negatively impact other firms with comparable production processes. When a firm expands to a new country with an unproven track record of dealing with foreign businesses, news about that firm's experience may affect other firms with plans to expand to that country. Consequently, the bellwether stocks do not have to be limited to the firms in the news and may comprise firms with high levels of investor attention, single-segment firms, or firms in the same supply chain, as suggested in previous literature.

In the number of news stories written about them, which, we argue, should be indicative of important news developments at the firm level, the number of stocks whose returns they lead increases as well. These findings are consistent with the notion that stocks may lead other stocks not only because they are quicker to react to common market or industry news but also because the firm itself may be at the origin of important news relevant for other firms.

There is ample evidence in the finance literature that prices react slowly to a firm's own news (e.g., the post-earnings announcement drift). Prices may react slower still when the relevant news is announced by another firm, especially when that news is of non-routine nature, which

makes it difficult to immediately assess its effect on the firm value. Every day, a large number of firms release new information. Using a near-complete sample of corporate press releases issued between April 2006 and August 2009, Neuhierl, Scherbina, and Schlusche (2013) document that, in total, about 218 valuation-relevant news are announced by firms each day. Only about 20% of these news items contain routine financial news (such as earnings, sales, dividends, plans to raise or return capital, etc.), while the rest make less routine announcements about products, partnerships, strategic plans, corporate lawsuits, and so on. These news announcements have the potential to affect valuations of other firms, but reaction time may be slow if investors overlook relevant news announcements by other firms due to limited attention or the inability to quickly assess the degree of relevance of other firms' news due to slow processing of complex information.

We show that news indeed travels slowly across stocks and that trading strategies can be devised to exploit delays in stock price reactions at monthly and weekly frequencies. We do not attempt to identify return leaders using ex-ante firm characteristics. Rather, we rely on the statistical ability of leader stocks to Granger-cause their followers' returns, which allows us to approach the question of how information flows across stocks purely empirically, without the need to first postulate the direction of the information flow. This approach, therefore, could help uncover new channels of information flow.

The methodology is implemented as follows. In every month (week) and for each combination of stocks i^* and j^* , we regress monthly (weekly) returns of stock i^* on the lag of its own return, the lag of stock j^* 's return, and the lag of the market return, using rolling regression windows of at least one year. Stock j^* is said to Granger-cause the return of stock i^* if the absolute value of the t -statistic on stock j^* 's lagged return exceeds 2.00 (or 2.57 in a robustness check). Having run these rolling regressions for all stock pairs, we are able to identify a set of leaders for each stock in each month (week), if such leaders exist. We hypothesize that the leaders' ability to forecast the returns of their followers will persist for at least another month (week). Hence, we proceed to calculate an aggregate predictive signal from all leaders for a follower's return. To calculate the aggregate leader signal for each follower, we first multiply the estimated regression coefficient on a leader's lagged return by the leader's current-

month's return to obtain that leader's signal and then compute the weighted average signal across all leaders of a particular follower stock. We confirm that this methodology is indeed able to identify lead-lag relationships that persist out-of-sample by showing that stocks with high aggregate leader signals earn high returns and stocks with low aggregate leader signals earn low returns in the subsequent month (week), controlling for other factors known to predict returns.

The leaders' ability to predict their followers' returns is unlikely to be explained by data snooping. To illustrate this, we scramble our panel data along the time dimension while preserving each cross section. The leaders that we identify using this scrambled dataset are, therefore, all false leaders that should not possess any predictive ability for their followers' returns, and we show that they indeed do not. In addition, we find that the return differential between the highest- and lowest-signal portfolios exhibits the properties of other documented anomalies; its magnitude declines over time and it is stronger for smaller, more neglected stocks. Finally, we show that short sellers increase their shorting demands for stocks that receive low leader signals, which suggests that sophisticated investors trade on leader signals.

We take steps to ensure that the predictive ability of leaders cannot be attributed to non-synchronous trading. We limit the sample of followers to only stocks that traded on the last day of the previous period, thus largely eliminating the concern about non-synchronous trading. In addition, in portfolio results, we require that all followers be priced above \$5 per share, which ensures that portfolios are comprised of rather liquid stocks. Moreover, the predictive ability of the leader signal survives skipping one month before portfolio formation for equal-weighted portfolios, and the weekly leader signal survives skipping up to three weeks for equal-weighted and up to two weeks for value-weighted portfolios.

Investors initially are more likely to underreact to small-firm news due to limited attention. We further illustrate that leaders may be small stocks by restricting the set of leaders to stocks that are smaller than their followers and showing that the strategy works almost as well.

Another important distinction from the lead-lag literature is that we are able to make within-industry long-short bets. In contrast, Hou (2007) documents that large firms in a particular

industry lead small firms in that industry, but not small firms in a different industry. Relying on this kind of large-firm signal would preclude making long-short bets within industries, as all stocks in the same industry will receive the same signal. Moreover, when we limit the set of leaders to stocks belonging to a different industry than the follower, the strategy still works. From an investor's perspective, an advantage of intra-industry long-short bets is that these result in industry-neutral long-short portfolios that are hedged against industry-wide shocks. Such portfolios have less volatile returns relative to portfolios sorted over the entire stock sample, since those will likely contain uneven industry loadings.

Recent papers have uncovered new channels of cross-firm information flows. In particular, Menzly and Ozbas (2010) document that information travels between supplier and customer industries and Hong, Torous, and Valkanov (2007) present evidence that some industries even have the ability to lead the entire market. Albuquerque, Ramadoraei, and Watugala (2015) show that information can flow slowly across countries by showing that firms with high trade credit located in producer countries have stock returns that are strongly predictable based on the returns of their associated customer countries. The information transfer literature in accounting shows that early earnings announcers predict earnings surprises of late announcers within the same industry. Again, these signals will be correlated for all followers within an industry, precluding within-industry long-short bets. Cohen and Lou (2012) show that information diffuses slowly from single-segment firms to multi-industry conglomerates. Cohen and Frazzini (2008) find that information travels slowly through the supply chain; in that setup, followers in the same industry may receive uncorrelated signals, but, similarly to the lead-lag literature, leaders tend to be larger firms.

These aforementioned papers assume that the set of leaders for a given firm is predetermined by the firm's customer-supplier ties or by the industry affiliation of its segments. An important advantage of the Granger causality methodology used in this paper is its ability to identify not only stable (long-term) leaders, such as those determined by supply-chain links, but also transitory (short-term) leaders, whose leadership for a given firm will disappear once the relevant news development is resolved, and which are, therefore, not identifiable through traditional data sources. Our paper is similar in spirit to Gatev, Goetzmann, and Rouwenhorst

(2006) in that both papers rely on a statistical technique to identify firm interconnections. Gatev, Goetzmann, and Rouwenhorst (2006) investigate the performance of a pairs trading strategy and, like us, document the strategy is profitable even after accounting for trading costs.

The paper proceeds as follows: Section II explains the methodology used to identify information leaders. Section III documents the ability of leaders to predict the returns of their followers out-of-sample. Section IV provides evidence that firms with a larger number of news stories written about them tend to have more stock followers. Section V concludes.

II. Identifying Information Leaders

We identify information leaders for each stock i^* based on their ability to Granger-cause stock i^* 's return. Specifically, using a rolling window of 12 months (or 36 months) including the current month τ , we run the following monthly regression for each combination of stocks i^* and j^* :

$$R_{it} = b_{0ij} + b_{1ij}R_{mkt,t-1} + b_{2ij}R_{i,t-1} + b_{3ij}R_{j,t-1} + \epsilon_{tij}, \quad (1)$$

where we require that both stocks i^* and j^* have 12 (36) monthly return observations available. Stock j^* is assumed to Granger-cause the return of firm i^* if the absolute value of the t -statistic for the estimated regression coefficient b_{3ij} is greater than 2.00 (or 2.57 in a robustness check). Furthermore, if the estimated coefficient b_{3ij} is positive, we say that stock j^* is a positive leader of stock i^* , and if negative, a negative leader.

When choosing the length of the estimation window, two considerations need to be balanced. On the one hand, it is beneficial to have a longer regression period to reduce noise. On the other hand, making the rolling window overly long will prevent us from uncovering relatively

short-lived leader-follower pairs. We therefore settle for two rolling window lengths, 12 months and 36 months.

Many leaders are misidentified as such due to estimation noise. The following quick calculation illustrates how many stocks are likely to be falsely identified as leaders for each stock i^* . For each potential follower i^* , the average number of cross-sectional regressions (1) being run every month equals the average size of the monthly cross section of stocks minus one for stock i^* itself, or $3,304.68 - 1$. Under the assumption that the distribution of the estimated coefficients $b^{3ij}b^{3ij}$ is perfectly normal, the associated likelihood of falsely identifying as leaders stocks whose true coefficient $b^{3ij}b^{3ij}$ equals zero is 4.55% (the two-tailed p-value corresponding to a t^* -statistic with an absolute value of 2.00). On average, this amounts to about 150 false leaders per follower.

Table 1 provides descriptive statistics for leaders and followers. The data are calculated as of January 31 of each year. Leaders are drawn from an unrestricted dataset that includes all stocks in the CRSP universe. We restrict the set of potential followers to domestic common stocks with share codes 10 or 11 that had a trade on the last day of the previous month and are priced at or above \$5 per share. Hence, leaders are drawn from a somewhat larger set of stocks than followers. The table shows that every stock eligible to be classified as a follower has, on average, 287 leaders (stock-month observations with no leaders are assigned a value of zero). This does not imply that the difference between 287 and 150 equals the number of independent leaders. Many "true" leaders, especially large leaders for small followers, are likely to offer correlated signals by virtue of reacting to common information shocks ahead of the followers. Hence, the number of "independent" leaders is likely to be smaller. Finally, a vast majority of firm-month observations have at least one leader.

When focusing on stocks that have at least one leader, the table shows that positive leaders slightly outnumber negative leaders. The absolute value of the coefficient $b^{3ij}b^{3ij}$ is about 0.9 for both positive and negative leaders. For a given follower, its leaders do not typically belong to the same industry, but more positive than negative leaders do. Despite the share price restriction on the followers and none on the leaders, the table shows that a follower stock tends to be smaller, to have lower turnover, and to be younger than its average leader stock.

The last sub-table sorts, every month, all followers into quintiles based on the number of leaders that a follower has. It can be seen that the stocks with the smallest number of leaders tend to be larger and more heavily traded than other stocks; this is consistent with the result from the lead-lag literature that smaller and less liquid stocks typically have more liquid, large-stock leaders, which are simply the first to react to common market- or industry-wide news. Of note, market capitalizations in that table appear small because they are averaged over the entire sample period from 1929 to 2011. Finally, while leader-follower relations exhibit some persistence over time, the probability that a leader-follower pair existed as such in the past declines smoothly when moving further back in time, which is likely reflective of the fact that the leader and the follower had fewer similarities further in the past (for more details, see Table A1 and Section A1 in the Online Appendix).

III. Return Predictability

Having obtained a set of $J_{\tau} I_{\tau i}$ leaders for each stock i^* in month τ , if such leaders exist, we proceed to calculate the aggregate leader signal as the weighted average of the products of the leaders' returns in month τ and the corresponding coefficient estimate b^3 :

$$\text{Signal}_{\tau i} = \sum_{j=1}^{J_{\tau i}} w_j b_{\tau i j}^3 \text{Ret}_{\tau j}, \quad (2) \quad \text{Signal}_{\tau i} = \sum_{j=1}^{J_{\tau i}} w_j b_{\tau i j}^3 \text{Ret}_{\tau j}, \quad (2)$$

where w_j is the weight on leader j^* 's signal. In our baseline set of results, signals are equal-weighted across stock i^* 's leaders, in which case $w_j = 1/J_{\tau i}$. The inset box in Figure 1 illustrates how the aggregate equal-weighted leader signal is computed.

In the following, we present results based on portfolio sorts and cross-sectional return regressions. The data used in the paper are described in Section A2 of the Online Appendix.

A. Monthly Portfolio Returns

1. Baseline Specification

In the baseline specification, we identify leaders with 12-month rolling regression windows and equal-weight signals across leaders. We compute signals for each follower stock in month τ and we sort followers into deciles based on the aggregate leader signal within each of the 36 industries that remain after the industry "Irrigation Systems" drops out and the stocks in the industry labeled "Other" are discarded. We form portfolios at the beginning of month $\tau+1$ and hold them for one month. In the following month, new portfolios are formed based on the new set of leader signals. Figure 1 illustrates the timeline for our regression window and portfolio formation.

Table 2 presents average monthly excess returns for various deciles of equal- and value-weighted follower portfolios (Panels A and B, respectively), along with return differentials between the highest- and lowest-signal portfolios. Over the 1929-2011 period, leaders possess significant out-of-sample predictive ability. Low-signal portfolios earn low returns and high-signal portfolios earn high returns, and returns increase smoothly in magnitude with the signal for both return-weighting methods. Moreover, alphas of the lowest-signal portfolio (decile 1) are significantly negative for both equal- and value-weighted returns, and the alphas for the highest-signal portfolio (decile 10) are significantly positive when equal-weighted, but not when value-weighted. The lack of significance of the value-weighted alpha on the highest-signal portfolio suggests that positive information is incorporated faster than negative information, at least for larger stocks. This observation is consistent with the evidence of Hong, Lim, and Stein (2000) that bad news diffuses more slowly than good news. The return differentials between the extreme decile portfolios are significantly greater than zero for both equal- and value-weighted portfolios and for all return measures (i.e., excess returns, alphas relative to the market, or three- or four-factor alphas). The raw return differentials are 0.61% with a t -statistic of 6.20, and 0.45% with a t -statistic of 3.39 for equal- and value-weighted portfolios, respectively. Since our portfolios are constructed to have the same industry loadings, industry-wide movements are canceled out for the return differentials, thereby reducing their volatility and increasing the Sharpe Ratio. The return differentials between highest and lowest leader-signal portfolios have positive loadings on size and book-to-market factors (see Panels A and B of Table A5 in the Online Appendix). Therefore, the three- and four-factor alphas of the return differentials are lower than the raw return differentials. The monthly four-factor alphas on the

return differentials are equal to 0.53% with a t -statistic of 5.12, and 0.38% with a t -statistic of 2.67, for equal- and value-weighted portfolios, respectively. The result that value-weighted portfolios generate lower return differentials than equal-weighted portfolios is consistent with the results of the lead-lag literature that large stocks, by virtue of receiving more attention from sophisticated investors who tend to hold large stocks, react faster to new information.

With regard to month-to-month portfolio transition probabilities, we find that there is some persistence in portfolio assignments in the next two months, with somewhat U-shaped transition probabilities, which indicates that the stocks in the high- and low-signal portfolios have a higher chance of remaining in their respective deciles than those with other portfolio assignments. However, this stickiness in the portfolio assignments disappears 12 months into the future. (These results are presented in Panel C of Table A5 in the Online Appendix.)

Panel C of Table 2 presents monthly portfolio returns for the specification in which leaders are identified using 36-month rolling regression windows. With a longer regression window, regression coefficients can be estimated more precisely, but there is a smaller chance of identifying short-lived leaders. As can be seen in the table, this methodology, on net, produces very similar returns to the baseline specification. Some differences in results between these two methods will be revealed in the robustness checks and the Fama-MacBeth cross-sectional regressions presented later in the paper.

Finally, in Panel D, we check how long it takes for the leader signals to be fully incorporated into their followers' prices. Specifically, we skip one month between the month in which the leader signals are computed and the month in which portfolios are formed. The four-factor alphas of the return differentials are no longer significant, either with a 12-month or with a 36-month rolling regression window. This result suggests that leader signals tend to be fully incorporated into followers' prices within one month.

In sum, the results show that leader signals have the power to predict their followers' returns in the next month. Moreover, the predictive ability of leaders works within industries, which is a result distinct from the lead-lag literature (e.g., Hou (2007)). We next show that the predictive ability of leaders is robust to various methods of aggregating leader signals.

2. Alternative Methods for Aggregating Leader Signals

Next, we try four alternative methods of aggregating leader signals. Unlike the baseline specification (2), these methods do not involve the magnitude of the estimated regression coefficient b^3 , but only its

sign: $Signal_{it} = \sum_{j=1}^J w_j sign(b^3_{ij}) Ret_{ij}$ $Signal_{it} = \sum_{j=1}^J w_j sign(b^3_{ij}) Ret_{ij}$. Throughout the paper, we will refer to the leader-return weighting methods that do not rely on the magnitude of b^3 as "non-parametric" weighting methods. Specifically, we use the following non-parametric leader return weighting methods: (1) equal-weighting; (2) weighting by the leaders' market capitalization; (3) weighting by the absolute t -statistic of b^3 ; and (4) weighting by the absolute value of b^3 .

The results are presented in Panel A of Table 3. A comparison with the results in Panels A and B of Table 2 shows that, for value-weighted portfolios, the original specification produces slightly larger return differentials than any of the four alternative methods, while, for equal-weighted portfolios, weighting by the absolute value of the t -statistics of b^3 produces the largest return differentials. Value-weighting leader returns produces the lowest return differentials, which suggests that signals from large leaders, which are overweighted in this weighting scheme, are incorporated by the followers faster than signals from small leaders, likely because large leaders are more visible.

Panel B of the table skips one month between the month in which leader signals are calculated and the month in which portfolios are formed, as in Panel D of Table 2. With the exception of the leader-return aggregation methodology in which leader returns are value-weighted, the ability of leaders to predict their followers' returns two months later can be observed for equal-weighted portfolios; for value-weighted portfolios, we do not observe leaders' predictive ability for any of the leader-return aggregation methods. However, when two months are skipped before portfolio construction, none of the return differentials are significant (results not shown).

The results in this subsection show that the predictive ability of leaders is robust to various methods of aggregating leader returns. Assigning larger weights to larger and, thus, more

visible leaders produces the worst return predictability. This result suggests that investor inattention may be contributing to the slow information diffusion from leaders to followers.

3. Smaller Leaders and Leaders from Other Industries

In this section, we assess the sensitivity of the results when various restrictions on the set of leader stocks are imposed. First, we restrict leaders to stocks that are smaller than their followers in order to show that information may flow from smaller to larger stocks and that leaders need not be large stocks. The results, presented in Panel A of Table 4, show that the return differentials are still positive and significant. For equal-weighted portfolios, the four-factor alpha of the return differential is 0.43% ($t^*=3.89$); for value-weighted portfolios, it is 0.28% ($t^*=2.02$). Thus, although the magnitudes are somewhat smaller than in the baseline specification, the results are still significant, indicating that small stocks can indeed lead larger stocks.

Next, we restrict leaders to stocks that are in a different industry than their followers. The results are reported in Panel B of Table 4. The return differentials are highly significant, with four-factor alphas of 0.49% ($t^*=4.47$) for equal-weighted portfolios and 0.40% ($t^*=2.99$) for value-weighted portfolios. These magnitudes are very similar to the baseline results, confirming that the predictive ability of leaders does not rely on same-industry links. This finding is important because it allows investors to construct industry-neutral long-short portfolios that are hedged against industry-wide shocks.

4. Placebo Test

One may become concerned that the return predictability documented here may be driven not by a lead-lag effect in stock returns, as we claim, but rather by some stock characteristic, such as idiosyncratic volatility, with stocks at the extremes of this characteristic behaving as though they have extreme leader signals. To address this concern, we argue that if the cross-sectional dimension of the data is preserved but the time series dimension used for identifying leaders is broken by scrambling the dataset along the time dimension, the signals from identified but, in this case, definitely false leaders will stop predicting their followers' returns.

Specifically, we preserve cross sections but scramble the data along the time dimension by assigning each time period a random number and then sorting the cross sections by this random number. We apply this algorithm to the monthly-frequency sample from January 1980 to December 2011; the sample is kept relatively short to ensure that the stocks exist for most of the considered time period. Since we require that both stocks in every potential leader-follower pair have returns in the prior 12 months, a relatively short sample helps ensure the maximum number of possible leader-follower pairs. Using the scrambled sample, we re-run regression (1) for all pairs of stocks that have return observations for the previous 12 months. As before, we select as leader-follower pairs those stocks for which $t\text{-statistic}(b_3) \geq 2.00$.

In 8.62% of the regressions, a leader is identified. This number is higher than the 4.55% p-value corresponding to a $t\text{-statistic}$ of 2.00 that one would observe when the distribution of the estimated coefficients is normal, implying that the actual distribution is fat-tailed. In 4.23% of the regressions, a positive leader is identified, and in 4.39%, a negative leader.

Next, we calculate equal-weighted leader signals according to equation (2) in each month τ and use them to predict followers' returns in month $\tau+1$ (note that we use followers' returns in the actual month $\tau+1$ and not the scrambled month $\tau+1$ in order to show that it is not the cross-sectional variation in some omitted stock characteristic observed in month τ that forecasts the cross section of the next month's returns). We form decile portfolios in month τ based on leader signals in that month and check portfolio returns in month $\tau+1$.

We obtain the following results. The average leader signal is equal to -2.87% for the lowest-signal decile and 3.13% for the highest-signal decile when signals are equal-weighted; these numbers are -2.24% and 2.80%, respectively, when signals are value-weighted. The raw return differentials between the extreme leader-signal deciles is 0.12% ($t\text{-statistic} = 0.71$) for equal-weighted portfolios and 0.31% ($t\text{-statistic} = 1.24$) for value-weighted portfolios. The corresponding four-factor alphas are 0.18% ($t\text{-statistic} = 0.98$) and 0.33% ($t\text{-statistic} = 1.27$) for equal- and value-weighted portfolios, respectively. Thus, our methodology does not produce any return predictability on the scrambled data.

When 36-month rolling windows are used to identify leaders, the results are similar. A leader is found in 5.55% of the regressions run (in this case, the distribution of the estimated coefficients b^3 is less fat-tailed than when using 12-month regression windows). In 2.95% of the regressions run, a positive leader is found, and in 2.60%, a negative leader is found. The average leader signal is -1.95% and 2.28% for the bottom- and top-decile portfolios, respectively, when signals are equal-weighted, and -1.67% and 1.90%, respectively, when value-weighted. The average raw return differential between the highest- and lowest-signal portfolios is -0.17% (t -statistic = -1.08) for equal-weighted portfolios and -0.05% (t -statistic = -0.19) for value-weighted portfolios. The corresponding four-factor alphas are -0.20% (t -statistic = -1.21) and -0.04% (t -statistic = -0.14), respectively. Again, we do not find any evidence of return predictability on the scrambled data.

Our inability to generate return predictability on the scrambled data is in contrast to leaders' significant ability to predict returns on the actual data for the same time period (see the results for equal-weighted portfolios over the same sample period reported in Table A3, Panels C and D, which correspond to 12-month and 36-month estimation windows, respectively). These results confirm that the success of the strategy hinges on identifying leaders that truly exhibit return leadership for their followers and on this leadership being, to some extent, preserved out of sample.

B. Weekly Portfolio Returns

As information technology gets cheaper, markets become more efficient. Consistently, the results reported in the Online Appendix show that the profitability of the monthly-frequency strategy of trading on leader signals has declined over time. As the market becomes more efficient in incorporating new information, switching to higher trading frequencies may be warranted. In this section, we explore the profitability of a weekly-frequency strategy of trading on leader signals, which should work well if the information from leader stocks is incorporated more quickly than with a one-month delay but more slowly than with a one-week delay. (Tellingly, the lead-lag literature uses weekly return frequencies to document the delayed price reaction of small stocks.) Additionally, higher frequencies will generate more data points, which

will allow us, in later sections, to study the interaction between leadership and news coverage as well as leadership and short selling (the news dataset is relatively short, starting in April 1996, and the short selling dataset is even shorter, starting only in July 2006).

The weekly portfolio construction methodology is similar to the monthly methodology. We run regression (1) with weekly returns using a 52-week rolling regression window. Weekly returns are computed as Monday-to-Friday returns, using the CRSP Daily Stock file, thereby aligning returns with the weekly factors obtained from Kenneth French's web site. Even though the window length is still about 12 months, we are able to estimate regression coefficients with greater precision. Once leaders are identified, we form portfolios every Monday using the equal-weighted aggregate leader signal from the previous week, computed as per equation (2), and hold stocks in the portfolios for one week. Portfolios are formed within each of the 36 industries, thus ensuring equal industry loadings in the long-short portfolios.

Table 5 presents the results. The return differentials between the highest- and lowest-signal deciles are strongly significant. For equal-weighted portfolios, the raw return differential is 0.46% per week ($t^*=11.88$), and the four-factor alpha is 0.47% ($t^*=12.21$). For value-weighted portfolios, the raw return differential is 0.27% ($t^*=5.79$), and the four-factor alpha is 0.28% ($t^*=6.14$). These results are economically large: an annualized four-factor alpha of 0.47% per week translates into about 27% per year (assuming 52 weeks, but compounding would be even higher). The strategy is thus highly profitable at weekly frequency.

We estimate break-even trading costs for the weekly-frequency trading strategy. For the simple high-minus-low leader-signal deciles in Table 5, the break-even trading costs are equal to 0.13% for equal-weighted portfolios and 0.08% for value-weighted portfolios. By way of comparison, using the TAQ dataset for the time period from January 1983 to August 2001, Sadka and Scherbina (2007) estimate the average effective spread for a typical stock and a typical trade to be 0.25%. Hedge funds are more skilled at minimizing trading costs than an average trader in the TAQ dataset, and their trading costs may easily fall below our estimated break-even values. The relatively low values of the estimated break-even trading costs for the simple trading strategy imply that the strategies trading on leader signals can support only small investment amounts since large amounts would entail large price impacts. Therefore, the dollar profitability

of trading on slow information diffusion that we document here is not very high. These results suggest that the market, though not perfectly efficient, is competitive in the sense that the arbitrage profits are small.

1. Alternative Signal Aggregation and the Speed of Information Diffusion

Next, as we did for monthly-frequency signals, we try alternative methods for aggregating weekly leader signals and check their forecasting ability for various lags between the week in which the signals are computed and the week in which portfolios are formed. The four alternative non-parametric methods of aggregating leader returns are described in Section A.2. Table 6 reports the results. The first row of the table shows that, as in the case of monthly-frequency signals, the baseline method works slightly better for value-weighted portfolios than the alternative methods. For equal-weighted portfolios, weighting leader returns by the absolute value of the t^* -statistics of b^3 works best; as before, value-weighting leader returns produces signals with the worst return predictability. In the next four rows of the table, we form portfolios based on leader signals lagged by one, two, three, and four weeks. This analysis shows that leader signals are fully incorporated into equal-weighted portfolios within the subsequent four weeks, and into value-weighted portfolios within the subsequent two to three weeks, depending on the specification.

2. The Interaction Between Leader Signals and Followers' Concurrent Returns

In this section, we test whether conditioning on the correlation between the leader signal and the follower's contemporaneous return improves the leader signals' predictive ability. We expect the predictive ability of the leader signal to be strongest among followers whose prices have not yet co-moved with a signal, as the efficacy of the signal depends on delays in the price responses of the followers. In fact, if a follower's price has already moved in the same direction as the signal this week, it will likely move in the opposite direction in the subsequent week due to the return reversal effect, which is strongly present at both monthly and weekly frequencies. In this case, conditioning future returns on the past leader signal may even become counterproductive.

Every week, within each of the 36 industries, all follower stocks are sorted into quintiles based on their leader signal and then, within each leader-signal quintile, into further quintiles based on their return in that week. The four-factor alphas of the subsequent week's portfolio returns, presented in Table 7, show that the leader-signal strategy works within each reversal quintile. As expected, the highest-leader-signal/lowest-prior-week return portfolio (portfolio 51) generates the highest return in the subsequent week, and the lowest-leader-signal/highest-prior-week-return portfolio (portfolio 15) generates the lowest return in the subsequent week. The four-factor alpha of the return differential between portfolios 51 and 15 is 1.48% per week (t -statistic=20.57) for equal-weighted portfolios and 0.96% per week (t -statistic=13.42) for value-weighted portfolios. These results show that the performance of the leader-signal strategy can be substantially improved by conditioning on the follower's concurrent return.

C. Cross-Sectional Regressions

We next run Fama-MacBeth cross-sectional regressions to test whether the aggregate leader signal has predictive power for future returns after controlling for other well-known predictors. The regressions are run at monthly and weekly frequencies. The dependent variable is the follower's return in the next month (week). The main independent variable is the aggregate leader signal computed at the end of the previous month (week). Control variables include size, book-to-market, momentum, short-term reversal, turnover, Amihud's illiquidity measure, idiosyncratic volatility, and the previous month's (week's) industry return.

Table 8 (monthly) and Table 8 (weekly, in the original) report the results. In all regression specifications and in all subsamples, the coefficient on the aggregate leader signal is highly statistically significant; it ranges in magnitude from 0.080 to 0.240 at monthly frequency. The highest coefficient estimate is obtained for the specification in which leaders are determined with 36-month regression windows. The reported range of the regression coefficients on the leader signal implies that if two otherwise identical stocks have leader signals that are different by, for example, 10 percentage points, then their next month's returns would differ by between 0.8 to 2.4 percentage points. As expected, the coefficient on the interaction between the leader signal and the previous month's return is negative and significant at the 10% level.

Regressions in Panel B include more controls (Compustat variables and daily return data, sample 1963-2011). The coefficients on the leader signal are somewhat lower than those in the longer sample, but nevertheless highly statistically significant across all specifications.

At weekly frequency (reported in the original paper's Table 8, not fully reproduced here due to space), the results confirm that the aggregate leader signal has independent predictive ability for followers' returns, and that the signal works best for followers with lower levels of investor attention (e.g., low institutional ownership, low analyst coverage, small size, low turnover). Monthly- and weekly-frequency leaders have independent predictive ability at weekly return horizons.

IV. Leadership and News Originating at the Firm

Thus far, we have established that leaders identified with Granger causality regressions can predict their followers' returns. In this section, we provide evidence that leaders may emerge not only because they are first to react to common market or industry news but also because important valuation-relevant news may originate at the level of the firm. Specifically, we show that a stock's return leadership is associated with newsworthy developments at the firm level. For that purpose, we again use the TRNA dataset, which is geared towards traders and investment professionals. The portion of the dataset made available to us by Thompson-Reuters includes only firm-specific English-language news about U.S.-based firms. This dataset is described in more detail in Section A5 of the Online Appendix. Since this dataset covers only US firms, for the purposes of the analysis in this section, we limit the set of potential leaders to common stocks with share codes 10 or 11. We use the first year of the TRNA sample to form the first annual cumulative news count, which reduces the regression sample to the period from April 1997 to December 2011.

A. Regression Specifications

We argue that return leadership arises, at least in part, because of news developments at leader firms and assume that the more important a news development experienced by a firm is, the more news stories about the firm will appear in the TRNA dataset. Of course, it is important to control for firm characteristics that affect news coverage. Incidentally, because the primary users of the TRNA dataset are investment professionals, the stock characteristics that lead to more detailed news coverage are also associated with sophisticated investor attention. In turn, as we know from prior literature, investor attention is associated with return leadership because it ensures that firms react to common market or industry news ahead of firms with lower levels of sophisticated investor attention. TRNA users will presumably demand more detailed coverage for large firms, firms with high turnover, and, perhaps, firms belonging to certain industries. Therefore, we include these and other firm characteristics that are associated with attention as well as industry or firm dummies in our regressions explaining the number of followers with the number of news stories. Although the TRNA dataset itself is primarily focused on firm-level news, we take additional care to count only news stories about firms when forming news counts (more detailed descriptions are provided in the next subsection).

In order to assess whether the number of a firm's followers is related to its news developments, we regress the number of followers that a firm has on the number of stories written about the firm's own news developments and a set of firm characteristics that are associated with sophisticated investor attention (e.g., institutional ownership, size, analyst coverage, turnover, industry assignment, past stock performance, book/market). Since leadership is determined over a one-year window, we use rolling one-year averages for all explanatory variables; for news, we calculate rolling total news counts over that window. We include year dummies because both the number of publicly traded stocks (and, hence, the number of followers) and news coverage change over the years.

The distribution of the number of followers for each stock in our dataset, which is computed using only end-of-year observations and which also includes stocks with zero followers, is plotted in Figure A2 of the Online Appendix. In Panel A, the monthly leadership specification with a 12-month rolling regression window is used, and in Panel B, the weekly leadership

specification with a 52-week rolling regression window is used. Since the number of followers is a count variable, the distributions are non-negative and right-skewed. The requirement that a potential follower traded on the last day of the week, which we impose at weekly frequencies, eliminates more stocks than the requirement that a potential follower traded on the last day of the month, which we impose at monthly frequencies. As a result, the average and median numbers of followers in Panel A (357.2 and 329, respectively) are greater than those in Panel B (299.9 and 269, respectively).

We use three regression specifications. In the first specification, we estimate our regressions using quasi-maximum likelihood (Poisson) which is an appropriate methodology for explaining a count variable. This estimation method produces consistent and asymptotically normal coefficient estimates even if the underlying distribution is not Poisson (Wooldridge, 2002). In the second regression specification, we also use the Poisson regression specification but include firm fixed effects (excluding industry dummies). This specification should mitigate concerns regarding omitted firm characteristics that may influence the number of followers and may be correlated with the control variables. In the third regression specification, we run Tobit regressions that treat the number of followers as a continuous variable but recognize that the number of followers cannot fall below zero. Regressions are run at weekly (monthly) frequencies for weekly- (monthly-) frequency leaders, and standard errors are clustered by firm in all regression specifications.

B. News Counts

News stories often start with a sequence of alerts informing readers in the headline about the topic of the forthcoming article while the article is being written. Once the article is posted, it may get updated, appended, overwritten, or corrected. Consequently, news items posted in TRNA are classified as either "Alert," "Article," "Append," or "Overwrite." All news items are further tagged with a news topic code.

A distinct news story for a given firm is assigned a unique identifier, the primary news access code (PNAC). This identifier allows the reader to keep track of a particular story unfolding with a

series of alerts and follow-up reports. Arguably, the more complex or significant a news development, the more items would appear under its assigned PNAC.

Since we like to show that news originating at the firm can give rise to information leadership, in the main regression specification, we make an effort to use a count that only considers news originating at a firm. Therefore, for our primary news count, we consider only stories tagged with a specific set of topic codes (listed in footnote 22 of the original paper) that indicate firm-centered events. We further require that the news be "impactful" — i.e., accompanied by abnormal turnover. We also construct five alternative news counts:

1. "Highly relevant corporate news": count only firm-centered news stories with a relevance score of 1 (indicating the firm is at the center of an event);
2. "All impactful news": count all news stories (excluding stories about trade order imbalances) that fall on abnormal turnover days;
3. "All highly relevant news": count all highly relevant news stories (relevance score = 1);
4. "All news": count all news stories written about a firm;
5. "Only new news": count only unique PNACs, thereby ignoring multiple alerts, appends, overwrites, and articles within one PNAC.

For firms with no news stories identified over a trailing one-year period when using a particular news count methodology and for firms not present in the TRNA dataset, we set the news count to zero. Because of the extreme right-skewness of the news count variables, we winsorize them at the 99th percentile. Summary statistics for the news count variables are reported in Panel D of Table 9.

C. Regression Results

The regression results are reported in Table 9. Pairwise correlations between the control variables are shown in Table A7 of the Online Appendix. Given that weekly-frequency leaders work best in forecasting their followers' returns over the time period of the news sample, we begin by reporting in Panel A the regression results explaining the number of followers at the weekly frequency. We use our primary news count, "impactful firm-level news." The results

show that the number of followers that a firm has is positively related to proxies for sophisticated investor attention, such as institutional ownership, analyst coverage, and, in one specification, share turnover; firm size is positively related to the number of followers only when other proxies for investor attention are not included. These findings are consistent with the lead-lag literature that stocks with higher levels of attention react faster to common shocks and therefore would appear to have more followers. Both analysts and institutional investors, who are sophisticated and react to new common information faster than retail investors, help uncover news developments relevant to a firm, thereby ensuring a quick price reaction to common news shocks. The negative regression coefficient on Momentum indicates that stocks that have experienced low returns tend to have more followers, consistent with the view that negative news travels slower across firms than good news.

Importantly, regression coefficients on the news count variables are significantly positive at the 1% level in all regression specifications. Poisson regression coefficients range from 0.0195×10^{-2} to 0.0432×10^{-2} . The economic interpretation is as follows: when a stock's news count increases from the 10th to the 95th percentile in the news count distribution, or from 0 to 117 news stories in the previous 12 months (Panel D), its number of followers increases by between 2.28% and 5.05% ($0.0195 \times 10^{-2} \times 117$ and $0.0432 \times 10^{-2} \times 117$, respectively). The median stock has 269 weekly-frequency followers; thus, the Poisson regression specification suggests that the median stock would gain between 6 and 14 new followers. Tobit regression coefficients range from 4.6856×10^{-2} to 12.1579×10^{-2} . These numbers suggest that the same increase in the news count is associated with an increase in the number of weekly-frequency followers between 5 and 14 ($4.6856 \times 10^{-2} \times 117$ and $12.1579 \times 10^{-2} \times 117$, respectively), which is close to the estimates produced by the Poisson regressions.

When the regressions are re-run to explain the number of monthly-frequency followers in Panel B, using, as before, our primary news count, the results are qualitatively similar. The Poisson regression coefficients in specifications (1)-(2) and (4)-(5) range

from 0.0149×10^{-2} to 0.0374×10^{-2} . These estimates imply that an increase in the number of news stories that would move the stock from 10th to the 95th percentile in the news distribution is associated with a 1.75% - 4.37% increase in the number of followers. The median stock has 329 monthly frequency followers (Figure A2 in the Online Appendix). Thus, these numbers would translate into 6 - 14 additional followers for a median stock. The Tobit regression coefficients range from 5.5819×10^{-2} to 13.8233×10^{-2} , which suggests that the same increase in the news count is associated with between 7 and 16 additional followers.

When the variable **News2News2** is included in the regression, the coefficient on that variable is negative, and significantly so in the regression explaining the number of monthly-frequency followers. This nonlinearity in the relation between the number of followers and the news count suggests that very intensive news coverage of a firm increases investor awareness of that leader's relevant news and, as a result, leads to shorter delays in followers' price reactions.

We substitute our primary news count with each of the five alternative news counts in Panel C, using regression specifications (4)-(6) of Panel A. (To conserve space, the Panel only reports the regression coefficients on the news counts.) The regression coefficients on all five alternative news counts are significantly positive in all regression specifications. As before, we compute how many additional followers are associated with a stock moving from the 10th to the 95th percentile of the distribution for each of the news counts that we have created, for both weekly- and monthly-frequency leaders. For the count of only "highly relevant corporate news," these numbers are 7 and 11, respectively; for "all impactful news," these numbers are 5 and 8; for "all highly relevant news," these numbers are 3 and 7; for both "all news" and "only new news," these numbers are 6 and 9. These results show that the relation between news and the number of followers is robust to various news counts. The question of what types of corporate news generate the largest increase in the number of followers is left to future research.

The results in this section show that important firm-specific developments are associated with a larger number of follower stocks. We take this to mean that being at the center of important news is associated with leadership. There could be alternative interpretations for this association. For example, one could argue that news coverage is another proxy for attention,

and, as we discussed, stocks with higher levels of attention are faster to react to common news; by this logic, such stocks will lead the returns of more followers who happen to react to the same news with a longer delay. However, this interpretation is unlikely because, firstly, the TRNA dataset itself focuses on firm-level news and, secondly, because we take additional efforts to ensure that we consider only firm-level news. Relatedly, one could argue that there might be an omitted variable that influences both leadership and coverage. News media dedicates more coverage to firms with a higher demand for coverage, that is, firms with higher institutional ownership and firms that trade more often. However, our results are still present when we control for stock characteristics that might be associated with a higher demand for coverage that influences both leadership and news coverage, and our results also hold in regressions with firm fixed effects. Still another alternative explanation for our findings is reverse causality: journalists pay more attention to firms that lead the returns of other firms. We consider this interpretation highly unlikely as it is not clear that Thompson-Reuters corporation keeps track of the statistical ability of stocks to Granger cause returns of other stocks and that this knowledge, in turn, influences the decision of how much coverage to allocate to a given firm.

To sum up, we believe the results presented in this section support our hypothesis that a firm's leadership is, to some extent, driven by news originating at the firm. Such news may have ramifications for other firms, but these firms' investors may initially overlook the relevant news, resulting in lead-lag return relations between the firms at the center of the news and other firms also affected by these news developments.

V. Conclusion

We use Granger causality tests to identify leader-follower pairs among individual stocks and show that the returns of leaders have a robust out-of-sample forecasting ability for their followers' returns that works within industries. Analyzing the uncovered leader-follower pairs, we find that information can flow in unexpected directions. In particular, we discover that

leaders can be smaller stocks. This finding gives rise to the conjecture that stocks can lead returns of other stocks not only because they are quicker to react to common market- or industry-wide news, but also because they might themselves be at the center of important news developments that have ramifications for other firms. We find support for this conjecture by showing that when stocks experience an increase in the news coverage of their firm-specific developments, the number of stocks whose returns they lead also increases. It is left to future research to investigate what types of firm-level news are most likely to affect other firms.

Our results suggest that individual stocks are highly interconnected by permanent or temporary business ties, similar legal liabilities, and similar exposure to product safety concerns, cross-investments, labor force regulations, consumer tastes, etc. Hence, it is natural that stock prices will react not only to own news and to market- and industry-wide news but also to relevant news of other firms. This observation can help resolve the R^2 puzzle articulated by Roll (1988), which states that asset pricing models do about as well in explaining individual stock returns on non-news days as they do on news days. Our results suggest that the difference in R^2 s between no-news and news days can be increased by adding to the set of news days the days on which a firm's return leaders experience significant news events.